

PROJECT DOCUMENTATION

AI FAQ ASSISTANT API

1. INTRODUCTION:

- Project Title: "Ai Faq Assistant Api"
- Team ID: NM2025TMID38548
- Team Leader: M. Divya sree
- Mail id: sreed1347@gmail.com [Coder]
- Team Members:
 - **NAME:** M.Anusree
 - **MAILID:** anusreemuruganandham2007@gmail.com [Coder]
 - **NAME:** N.Deepika
 - **MAIL ID:** deepikapooja2728@gmail.com [Editing]
 - **NAME:** M.Dhivya sree
 - **MAIL ID:** dhivyaleela30@gmail.com [Documenter]
 - **NAME:** A.Ambeeleeswari
 - **MAIL ID:** ambeeleeswari14@gmail.com [Documenter]

2. PROJECT OVERVIEW:

- **Purpose:**
 - ❖ API-based platform for managing FAQs and providing AI-powered answers to user queries.
- **Features:**
 - ✓ AI-powered question answering
 - ✓ FAQ creation & management
 - ✓ Natural language query processing
 - ✓ Knowledge base search & retrieval
 - ✓ REST API integration

PROJECT DOCUMENTATION

3. ARCHITECTURE:

Frontend (What the user sees):

- The part of the application where users interact with the AI FAQ Assistant.
- Provides a simple interface for entering questions and viewing AI-generated answers.
- Built using **React.js, Bootstrap, and Material UI**.

Backend (Logic & Server-side):

- The core part of the application that receives user questions, processes requests, and connects with the AI service.
- Handles API requests and manages communication between the frontend, AI model, and database.
- Developed using **Node.js and Express.js**.

AI & API (Intelligent Processing):

- Understands user questions and generates relevant answers based on the available FAQ information.
- Provides AI-powered responses through an API.
- Implemented using an **AI API** integrated with the backend.

Database (Data Storage):

- Stores FAQ data, user information, questions, and responses for retrieval when needed.
- Implemented using **MongoDB**.

4. SETUP INSTRUCTION:

Prerequisites:

Node.js , MongoDB ,Git ,React.js ,Express.js ,VS Code , AI API, Web Browser

5. INSTALLATION:

1. Clone repo → `git clone <repo-url>`

2. Client → `cd client && npm install`

PROJECT DOCUMENTATION

3. Server → `cd ../server && npm install`

6. FOLDER STRUCTURE:

```
├── client/           (frontend)
│   ├── components/
│   └── pages/
│
├── server/           (backend)
│   ├── routes/
│   ├── controllers/
│   ├── models/
│   └── middleware/
│
├── .env
├── package.json
└── README.md
```

7. RUNNING THE APPLICATION:

- ❖ **Frontend** → `cd client && npm start`
- ❖ **Backend** → `cd server && npm start`
- ❖ **Access** → `http://localhost:3000`

8. API DOCUMENTATION:

Users: `/api/user/register`, `/api/user/login`

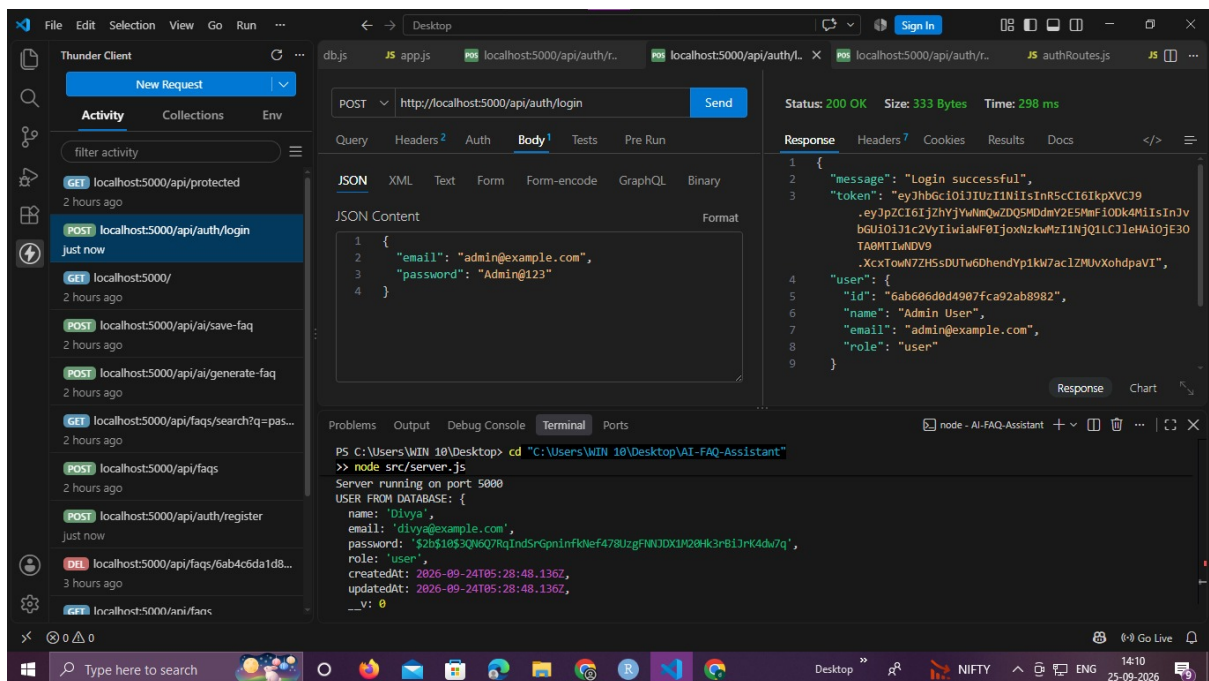
FAQs: `/api/faq/create`, `/api/faq/:id`

Questions: `/api/questions/ask`, `/api/questions/:id`

AI Assistant: `/api/ai/ask`, `/api/ai/response`

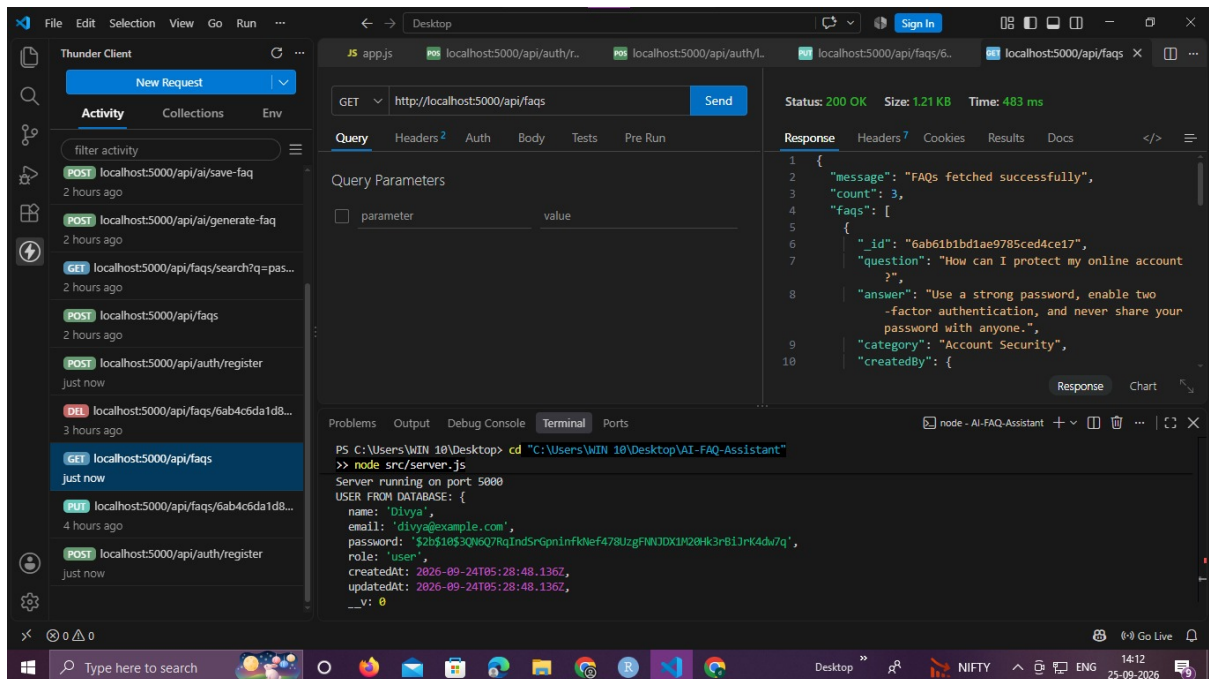
Admin: `/api/admin/faqs`, `/api/admin/users`

REGISTER :

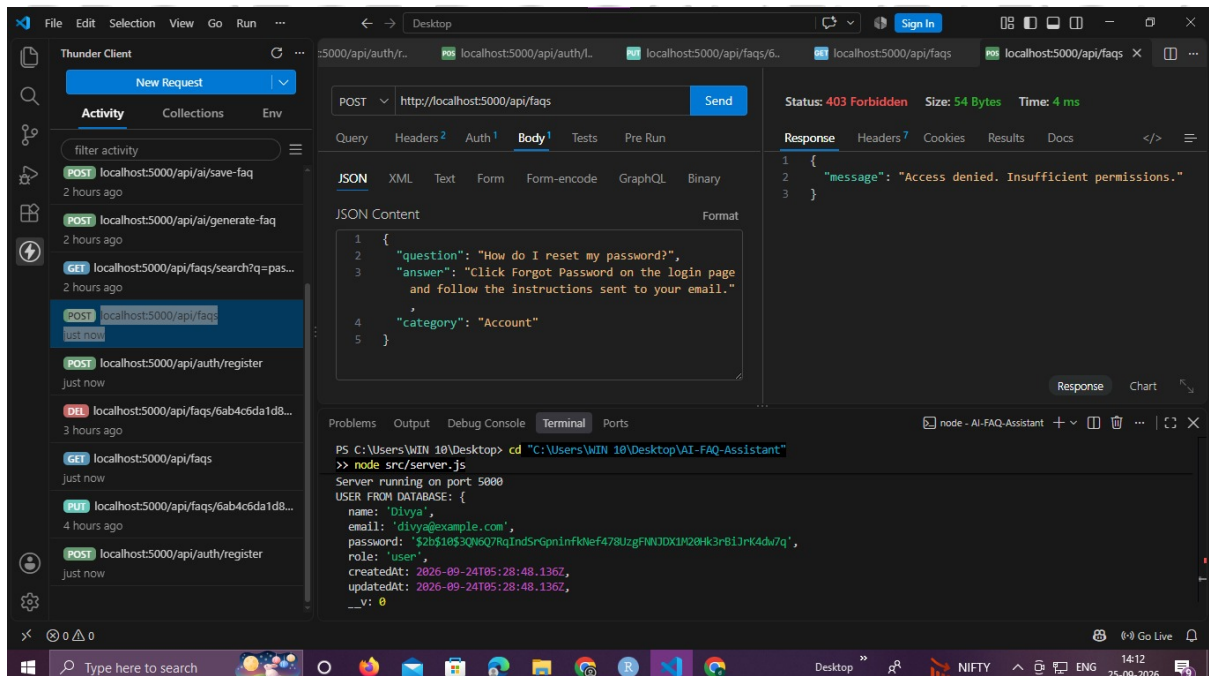


PROJECT DOCUMENTATION

API FAQ GET:

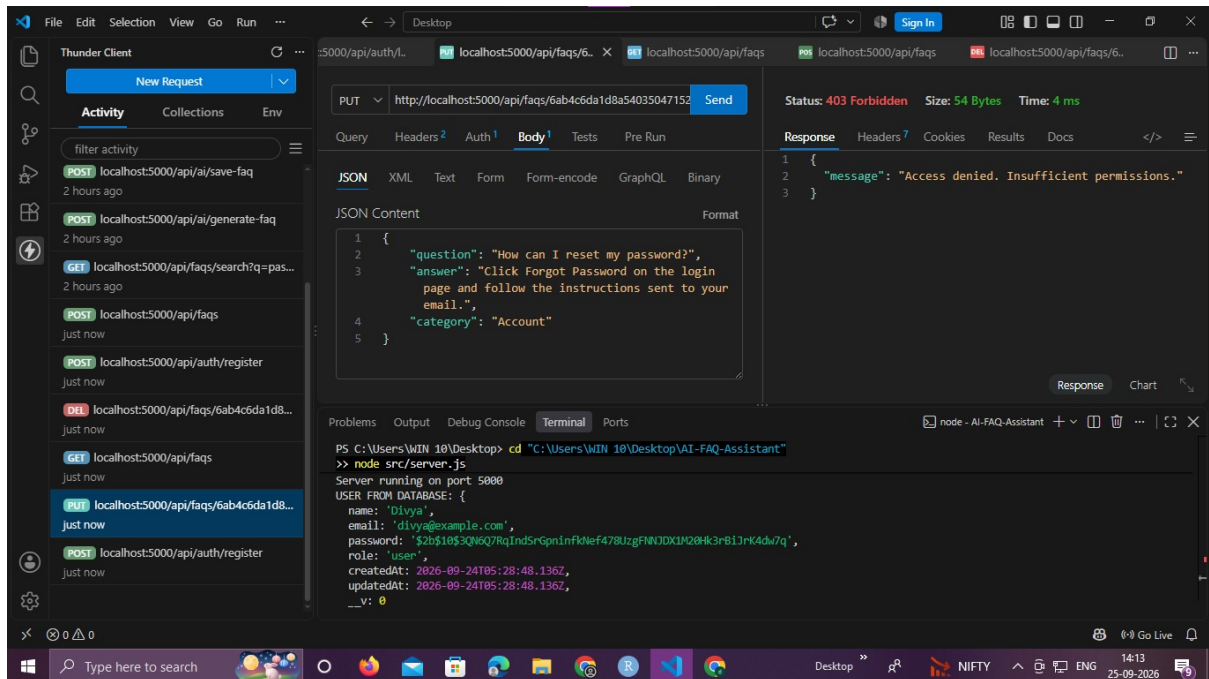


API FAQ POST :

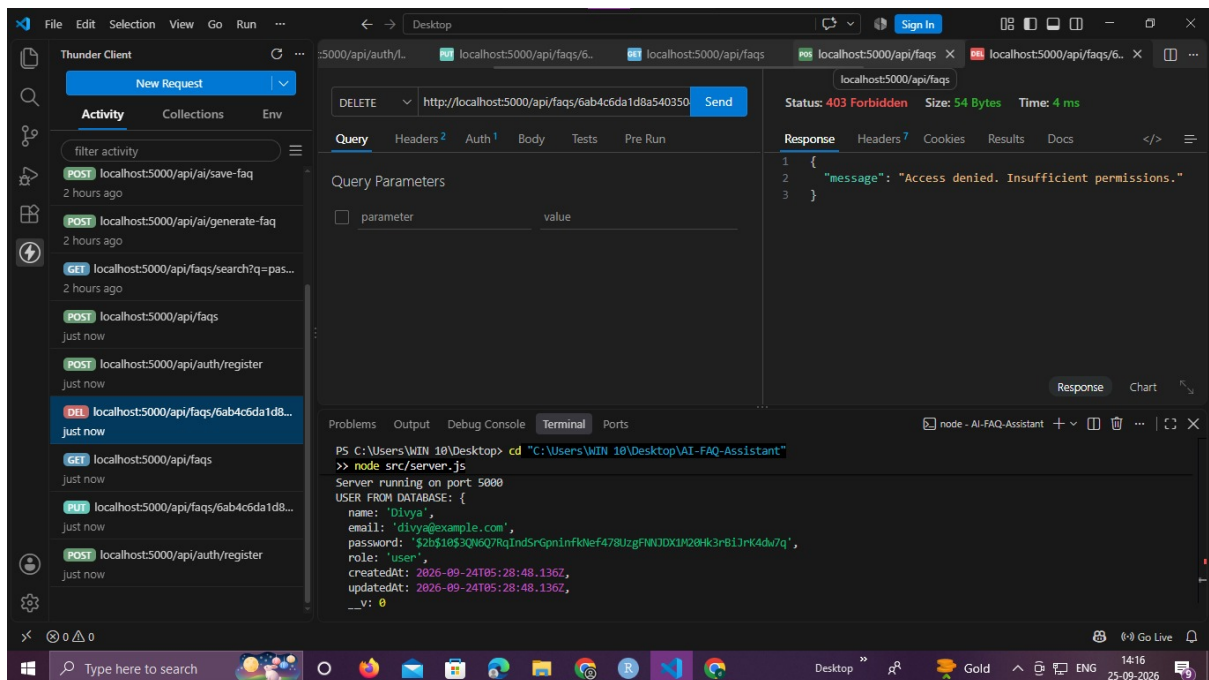


ID PUT :

PROJECT DOCUMENTATION

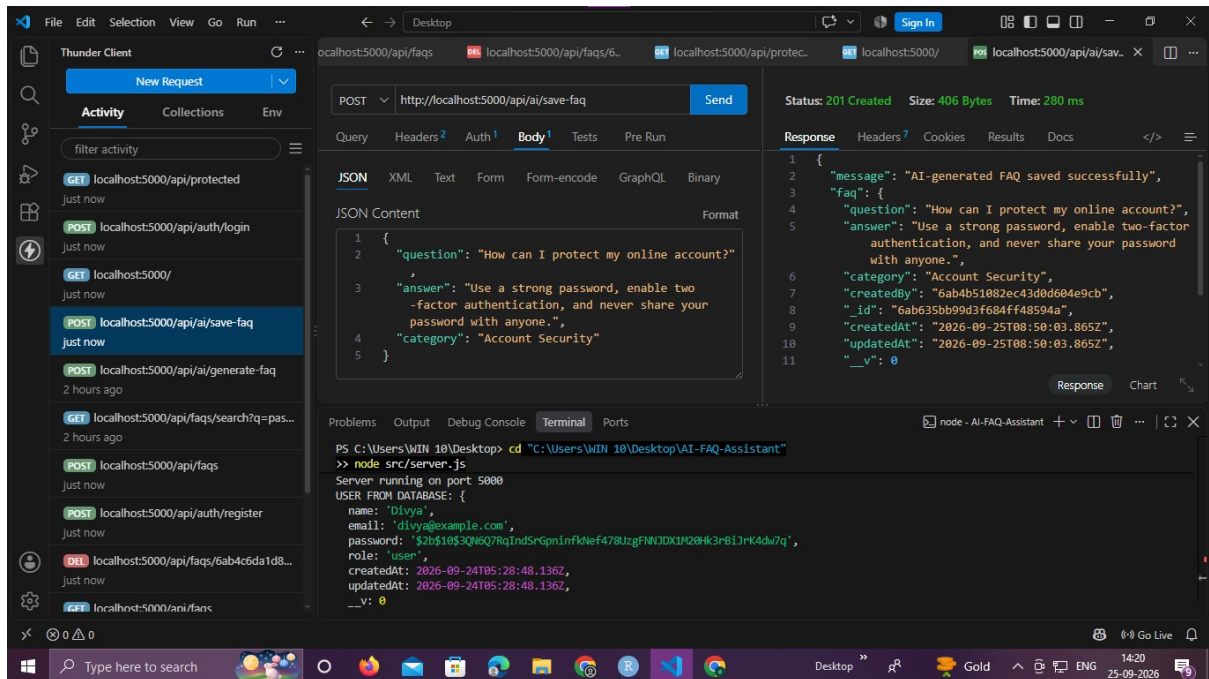


ID DELETE :

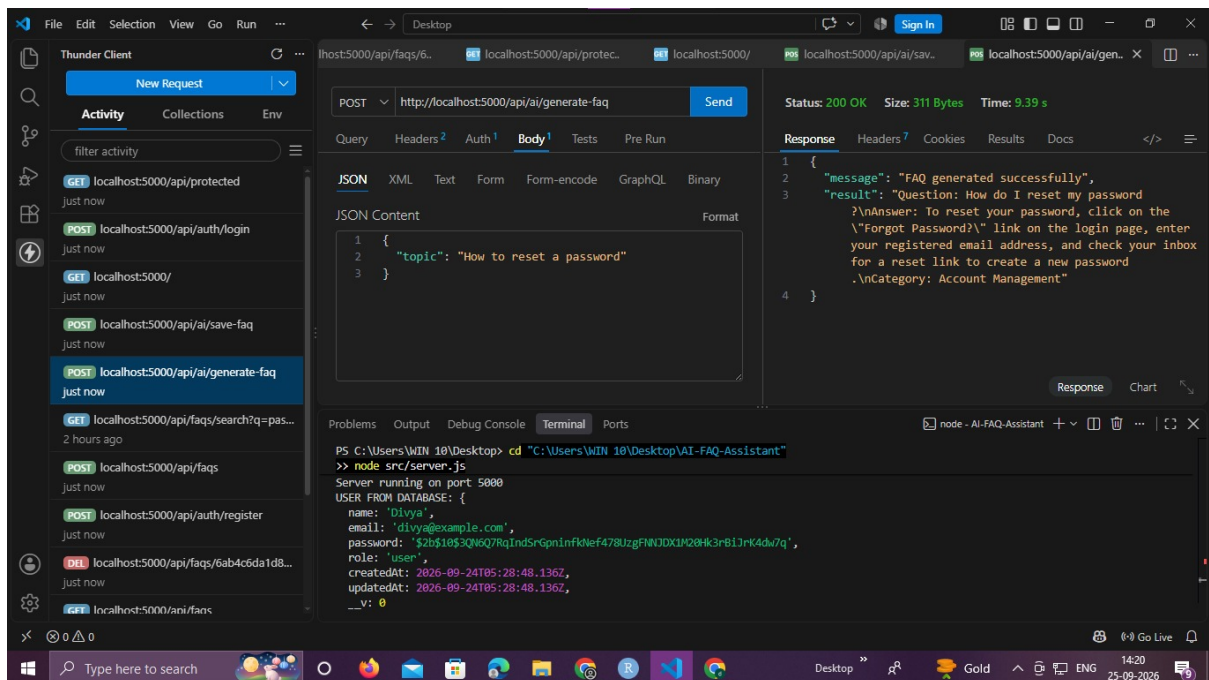


SAVE FAQ POST :

PROJECT DOCUMENTATION

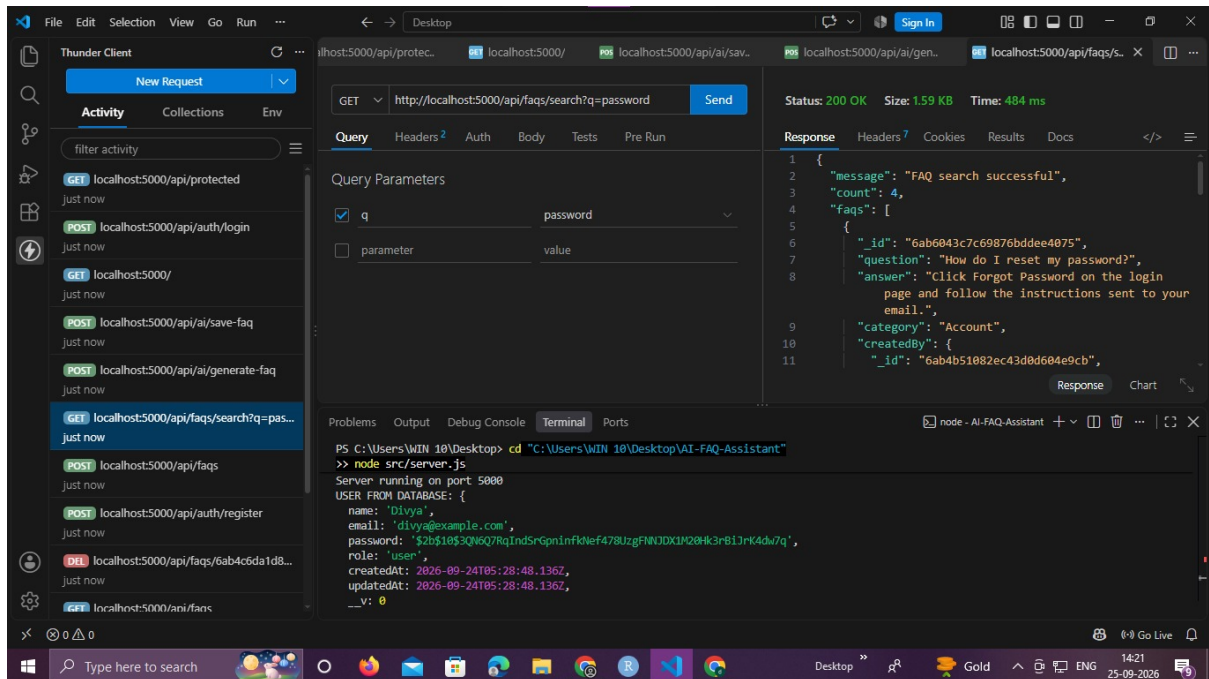


GENERATE FAQ POST :

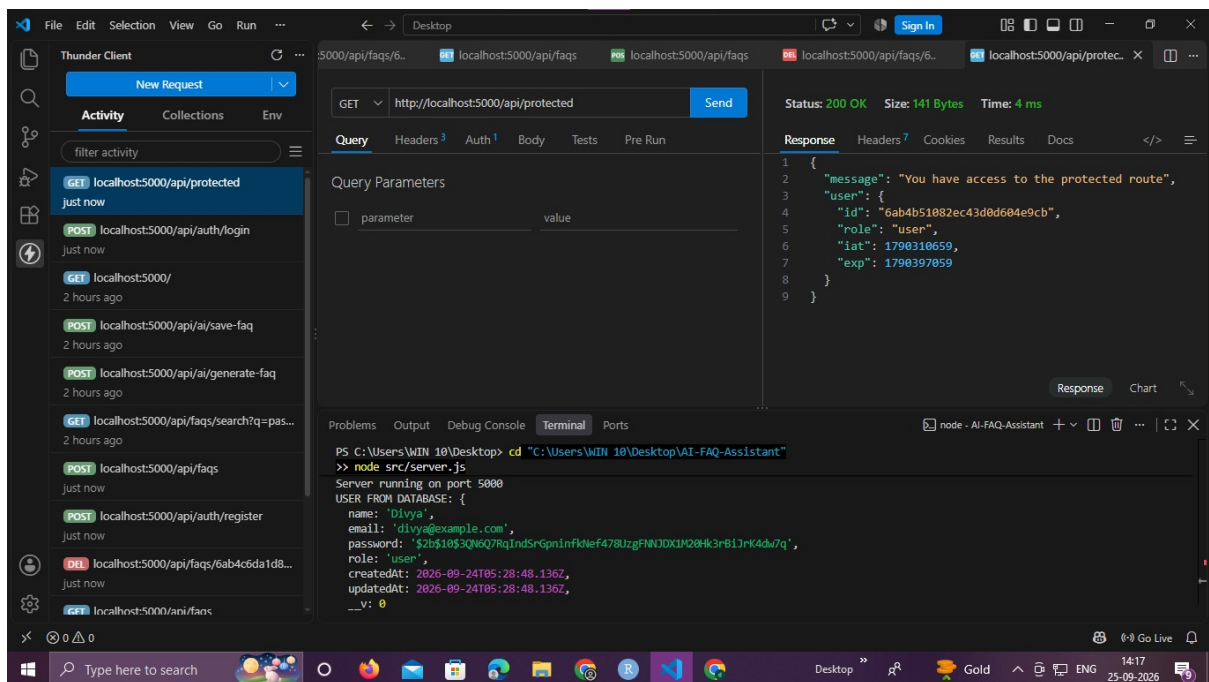


SEARCH FAQ GET :

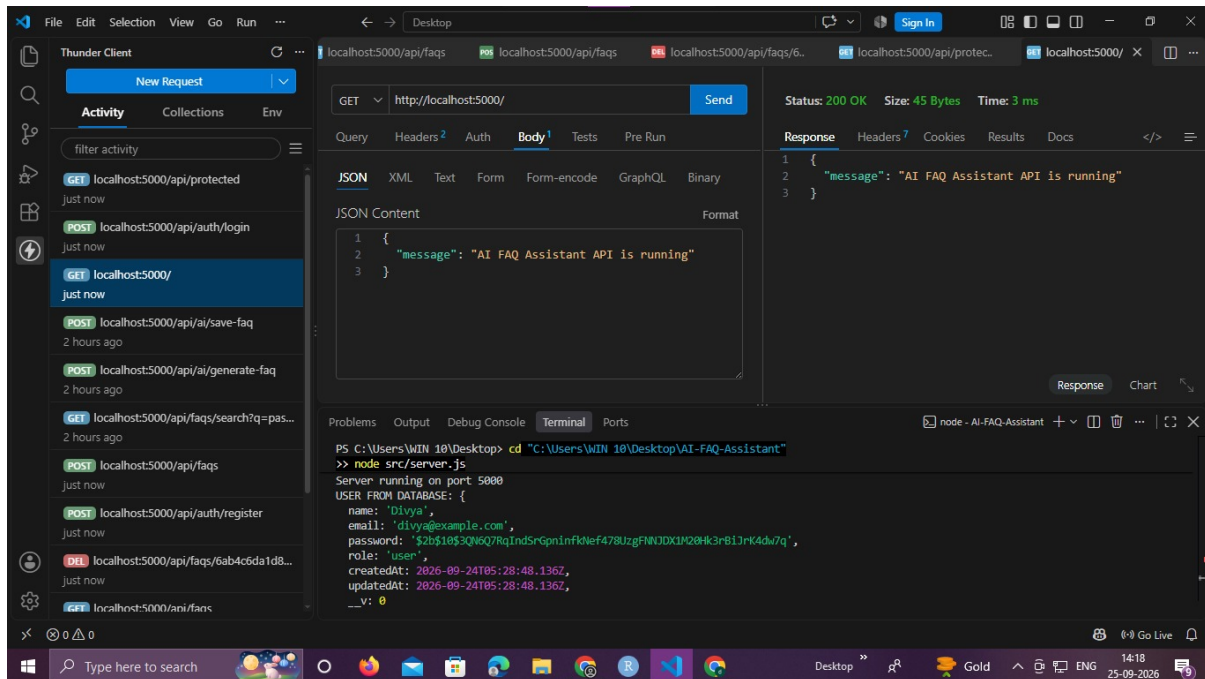
PROJECT DOCUMENTATION



API FAQ PROTECTED :



PROJECT DOCUMENTATION



11. KNOWN ISSUES

- Limited Natural Language Understanding
- FAQ Data Accuracy & Consistency

12. FUTURE ENHANCEMENTS:

- ❖ Voice-based FAQ Interaction
- ❖ Multi-language Support
- ❖ Advanced Search and Question Filtering

13. SYSTEM WORKING :

- The application receives a question from the user through the frontend. The request is sent to the backend, which processes the question and communicates with the AI service and database to produce a suitable response.
- User enters a question.
- Frontend sends the request to the backend.

PROJECT DOCUMENTATION

- Backend processes the request.
- AI service generates the response.
- Answer is displayed to the user.

14. SYSTEM BENEFITS :

- ✓ The AI FAQ Assistant provides a convenient way to access frequently required information. It combines automated responses with structured FAQ management to make information access faster and easier.
- Saves time
- Reduces repeated manual searches
- Provides quick answers
- Centralizes FAQ information
- Supports future improvements

15. SECURITY AND DATA HANDLING :

- The system manages user and FAQ information through the backend and database. User-related operations such as registration and login help control access to the application.
- User authentication
- Database management
- Controlled API access
- Organized data storage

16. PRACTICAL APPLICATION AND USABILITY :

- The AI FAQ Assistant API provides a practical solution for handling commonly asked questions through an automated system. Instead of depending

PROJECT DOCUMENTATION

completely on manual support, users can interact with the application and obtain information through a structured question-and-answer process. The system connects the user interface, backend services, database, and AI functionality to manage requests efficiently. This approach can be useful in applications where users require quick access to repeated or frequently requested information.

Key Points:

- Provides automated assistance for common questions.
- Makes information easier to access and understand.
- Reduces repetitive work in handling frequently asked queries.
- Maintains FAQ-related information in an organized manner.
- Provides a foundation for further development of intelligent support systems.

17. RELIABILITY AND MAINTANABILTY :

- The application is organized into separate components so that different functions can be managed independently. The frontend handles user interaction, while the backend manages requests and application logic. Database services maintain the required information, and the AI component supports response generation. This separation makes the system easier to maintain, test, and improve when additional requirements are introduced.

Key Points:

- Modular application structure.
- Easier maintenance and testing.
- Separate handling of user interface and backend operations.
- Organized storage of application information.
- Supports future modifications and feature expansion.

PROJECT DOCUMENTATION

18. ADVANTAGES OF THE SYSTEM

- ❖ The AI FAQ Assistant API provides an efficient way to handle frequently asked questions through an automated question-and-answer system. It reduces the need for users to manually search through large amounts of FAQ information and allows them to obtain responses through a simple interface. The integration of AI with the API also makes the system more interactive and useful for handling user queries.

Major Advantages:

- **Quick Information Access:** Users can obtain answers without searching through the complete FAQ collection.
- **Automated Responses:** The system can generate responses with the support of AI.
- **Easy FAQ Management:** FAQ information can be created, updated, retrieved, and deleted through API operations.
- **Reduced Manual Work:** Repeated questions can be handled automatically.
- **Centralized Information:** FAQ data can be maintained in one database.
- **Scalable Structure:** The system can be extended with additional FAQs and features.
- **User-Friendly Interaction:** Users can communicate with the system through a simple web interface.

19. DISADVANTAGES OF THE SYSTEM

- ✓ Although the AI FAQ Assistant provides automated support, it also has certain limitations. The quality of the response depends on the available FAQ information and the AI processing. If the required information is not available or a question is unclear, the system may not always provide the expected response.

PROJECT DOCUMENTATION

Major Disadvantages:

- **Dependence on AI Service:** AI-based responses depend on the availability of the integrated AI service.
- **Response Accuracy:** Some questions may not receive completely accurate or relevant answers.
- **Limited Knowledge Base:** The system can provide better results when the FAQ information is properly maintained.
- **Internet Dependency:** API and AI communication may require a stable network connection.
- **Maintenance Requirement:** FAQ information needs to be updated regularly.
- **Complex Questions:** Questions outside the available information may be difficult for the system to handle effectively.
- **Data Management:** User and FAQ information must be properly maintained to ensure reliable system operation.

OVERALL COMPARISON

The AI FAQ Assistant offers advantages in terms of automation, accessibility, and efficient FAQ management. At the same time, its effectiveness depends on the quality of the stored information, AI service, and proper system maintenance. Therefore, regular updating of the FAQ data and continuous improvement of the application can help address its limitations.

PROJECT DOCUMENTATION

CONCLUSION :

- ✓ The **AI FAQ Assistant API** successfully provides an automated approach for managing frequently asked questions and responding to user queries. By combining a React.js frontend, Node.js and Express.js backend, MongoDB database, and AI API, the system creates a complete flow for submitting questions, processing information, and delivering responses.
- ✓ The project demonstrates the practical use of AI and REST API technologies in developing an interactive information-support system. It simplifies access to FAQ information, reduces repetitive manual work, and provides a structured platform that can be maintained and extended with additional features in the future.
- ✓ The system also provides a clear separation between different application components, making the overall implementation easier to understand and manage. The API-based structure allows the frontend to communicate with backend services while the database maintains the required application information.
- ✓ Overall, the project demonstrates how artificial intelligence can be combined with modern web development technologies to create a useful FAQ management and question-answering solution. The developed system provides a foundation for building more advanced AI-assisted applications with improved capabilities and broader functionality.

PROJECT DOCUMENTATION

DEMO VIDEO :

DEMO LINK :

https://drive.google.com/file/d/1pcNfOwUBFRRjTzKxwbNx6KoZJPc3d6_a/view?usp=drivesdk

API LINK :

<https://drive.google.com/file/d/1mCbxKOfyjbV5CleI5pw7IJzHD8QAZZ/view?usp=drivesdk>

PROJECT DOCUMENTATION